

Hypothesis testing with e-values

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Chapters 1–2

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- A **p-value** p is the observed value of a nonnegative statistic (**p-variable**) P
- Under H_0 , P is **probability-constrained**: $\Pr(P \leq \alpha) \leq \alpha$
- Smaller p suggests H_0 is false
- Testing: reject H_0 at level α if $p \leq \alpha$.

- An **e-value** e is the observed value of a nonnegative statistic (**e-variable**) E
- Under H_0 , E is **expectation-constrained**: $\mathbb{E}[E] \leq 1$
- Interpret e as *evidence against* H_0 ; larger e means stronger evidence.
- Testing: reject at level α if $E \geq 1/\alpha$

Many p's, many e's...

p p-value

P p-variable

$\mathbb{P}$ Distribution in the null $\mathcal{P}$

$\mathcal{P}$ Set of probability distributions (Null)

e e-value

E e-variable

e exponential

Example

- $Y_i \sim \mathcal{N}(\mu, 1)$
- Test $H_0 : \mu = 0$ (vs $H_1 : \mu > 0$)
- Let $Z_n = \sqrt{n}\bar{Y}_n$
- p-variables

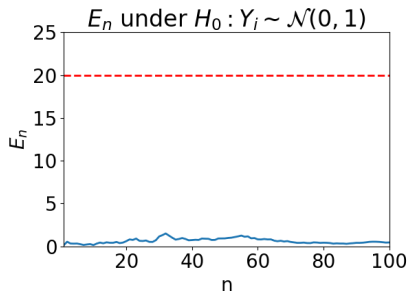
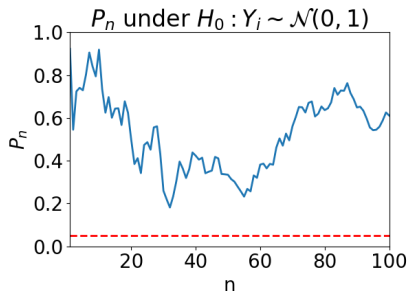
$$P_n = \Phi(-Z_n)$$

- e-variables

$$E_n = e^{Z_n - 1/2}$$

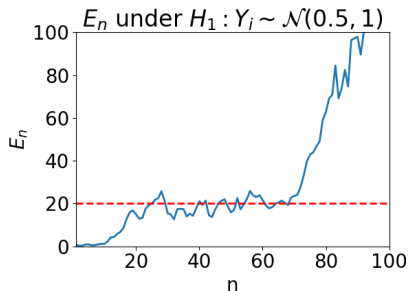
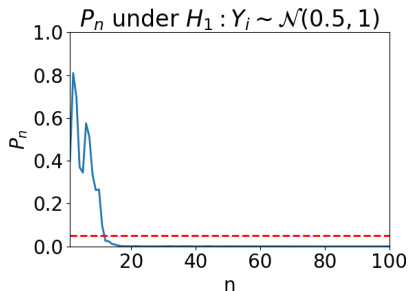
Example

- Under $H_0 : Y_i \sim \mathcal{N}(0, 1)$



Example

- Under $H_1 : Y_i \sim \mathcal{N}(0.5, 1)$



Some limitations/misuses of p-values

- Type-I control valid for a **fixed** α and **fixed** n
- For any $\alpha \in (0, 1)$, $n \geq 1$, under H_0 ,

$$\Pr(P_n \leq \alpha) \leq \alpha$$

or equivalently

$$\mathbb{E} \left[\frac{1_{P_n \leq \alpha}}{\alpha} \right] \leq 1.$$

- Does not hold in general if α or n depends on the data

Some limitations/misuses of p-values

- α -hacking/roving α /moving the goalposts
 - Wanted to use $\alpha = 0.05$
 - Observe $p = 0.053$; sure this is still quite small
 - ✗ Decide to set α to 0.01
- Wanted to use $\alpha = 0.05$
- Observe $p = 10^{-9}$; surely there is strong evidence against H_0
- ✗ Adjust α to 0.001

Some limitations/misuses of p-values

- n -hacking/optional stopping

- Run experiment with $n = 100$

- Evaluate $p_{100} = 0.07 > \alpha$

- ✗ Decide to collect another 100 samples and reject if $p_{200} < \alpha$

- Experiments are very costly; run them sequentially

- ✗ For $n = 1 \dots, 100$, reject and stop the experiment whenever $p_n \leq \alpha$; otherwise indecisive

Some limitations/misuses of p-values

- n -hacking, optional stopping (another version)
 - Collection of p-variables P_{n,θ_0} to test $H_{0,\theta_0} : \mu = \theta_0$
 - Inverse to obtain $1 - \alpha$ confidence set

$$C_{\alpha,n} = \{\theta_0 \mid P_{n,\theta_0} > \alpha\}$$

- For fixed α , n , $1 - \alpha$ confidence set

$$\Pr(\theta_0 \in C_{\alpha,n}) \geq 1 - \alpha$$

- Not valid anymore under optional stopping
- ✗ E.g: Stop as soon as the size of the set is below 1

Some limitations/misuses of p-values

- Consider multiple p-variables $P_1, \dots, P_k$
- Merging p-variables is generally nontrivial
- Similarly, combining confidence sets is nontrivial

p-values and e-values are statistical *cousins*

- They exist under the same conditions; each can be transformed to the other via *calibration* (Ch. 2.3).
- Neither dominates universally; they have *complementary strengths*
- For any p-variable P , $E = 1_{P \leq \alpha} / \alpha$ is an e-value
 - For testing, truncating P at α is the same as truncating $1_{P \leq \alpha} / \alpha$ at $1/\alpha$
- For any e-variable E , $P = \min(1, 1/E)$ is a p-value
 - For testing, truncating E at $1/\alpha$ is the same as truncating $\min(1, 1/E)$ at α

p-values and e-values are statistical *cousins*

- E-values play three roles:
 - 1 **Methodological:** universal inference; post-hoc decisions; optional continuation; unknown sampling schemes; safe anytime-valid inference; confidence sequences.
 - 2 **Technical:** tools to build/understand other procedures (e.g., compound e-values).
 - 3 **Fundamental:** core objects for anytime-valid inference and decision making.

e-values as (generalized) likelihood ratios

- For simple hypothesis, e-values are **likelihood ratios** (or are dominated by LRs)
- Sketch: Exact e-variable $E = e(X)$ with continuous data X with density $p(x) > 0$

$$\mathbb{E}[E] = \int e(x)p(x)dx = 1$$

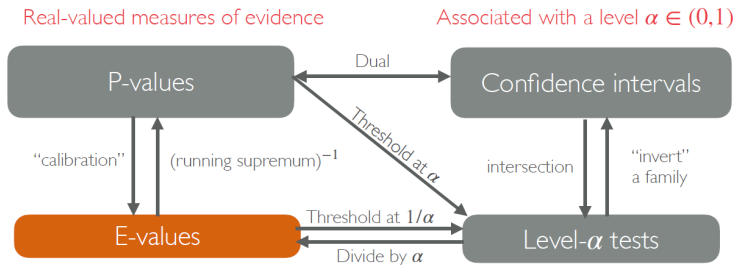
- $e(x)p(x) = q(x)$ is a density and $e(x) = q(x)/p(x)$ is a LR
- In composite settings, e-values can be thought as generalized LRs

e-values as (generalized) likelihood ratios

- $Y_i \sim \mathcal{N}(\mu, 1)$
- Test $H_0 : \mu = 0$ (vs $H_1 : \mu > 0$)
- Let $Z_n = \sqrt{n}\bar{Y}_n$
- e-variables

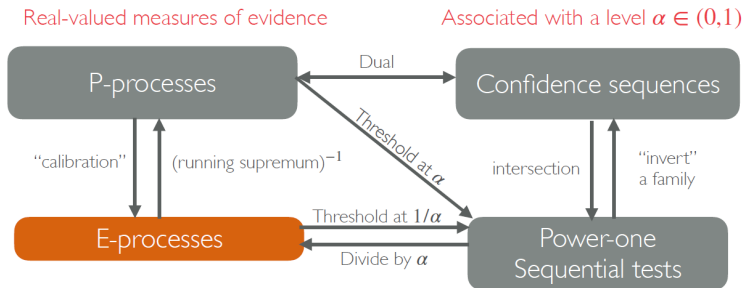
$$\begin{aligned} E_n &= e^{Z_n - 1/2} \\ &= e^{-\frac{1}{2}(Z_n - 1)^2 - \frac{1}{2}Z_n^2} \\ &= \frac{\mathcal{N}(Z_n; 1, 1)}{\mathcal{N}(Z_n; 0, 1)} \\ &= \frac{\prod_{i=1}^n \mathcal{N}(Y_i; 1, 1)}{\prod_{i=1}^n \mathcal{N}(Y_i; 0, 1)} \end{aligned}$$

Non-sequential



(From A. Ramdas)

Sequential Anytime-valid inference



(From A. Ramdas)

Core definitions

- $\mathcal{M}_1$ is the set of all probability measures on $(\Omega, \mathcal{F})$.
- $\mathcal{P} \subseteq \mathcal{M}_1$ be the null.
 - If $\mathcal{P} = \{\mathbb{P}\}$: **simple** hypothesis; otherwise **composite**
 - Null may be parametric or nonparametric
- $\mathcal{Q} \subseteq \mathcal{M}_1$ be the alternative (may also be simple or composite).

Core definitions

Let $\mathcal{P}$ be the null.

E-variable, p-variable, test

- **e-variable:** $E \in [0, \infty]$ with $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$; *exact* if $\mathbb{E}^{\mathbb{P}}[E] = 1$ for all $\mathbb{P} \in \mathcal{P}$.
- **p-variable:** $P \in [0, \infty)$ with $\mathbb{P}(P \leq \alpha) \leq \alpha$ for all $\alpha \in (0, 1)$ and $\mathbb{P} \in \mathcal{P}$ (often truncated to $[0, 1]$); *exact* if $\mathbb{P}(P \leq \alpha) = \alpha$.
- **test:** $\phi \in [0, 1]$ with type-I error $\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}^{\mathbb{P}}[\phi]$; *level- α* if this is $\leq \alpha$.
- Useful facts: $F_X(X)$ is a p-variable; arithmetic averages/convex combinations of e-variables are e-variables; products are e-variables if independent.

Betting interpretation

- View $E(\omega)$ as the payoff of a £1 bet against $\mathcal{P}$
- Skeptic pays £1 to Forecaster; receives $E(\omega)$ in return
- Forecaster believes that $\mathcal{P}$ describes well the world; they accept if it is an e-variable (valid bet: nonpositive expected loss).
- Skeptic believes that $\mathbb{Q}$ describes the world better than $\mathcal{P}$ and chooses E so that $\mathbb{E}^{\mathbb{Q}}[E] > 1$

Chapter 2: Validity under the null

Focus: how to test with e-values; how to convert to/from p-values; admissibility; examples.

2.1 Testing with Markov's inequality

Markov for e-values

If E is an e-variable and $\alpha \in (0, 1]$, then

$$\mathbb{P}(E \geq 1/\alpha) \leq \alpha \mathbb{E}^{\mathbb{P}}[E] \leq \alpha \quad \text{for all } \mathbb{P} \in \mathcal{P}.$$

Hence $1/E$ is a p-variable and $1\{E \geq 1/\alpha\}$ is a level- α (binary) test.

- p-variable $1/E$ not uniform and conservative
- Type-I error of the test $1\{E \geq 1/\alpha\}$ usually much smaller than α
- For a p-variable P and e-variable E , tests $1\{E \geq 1/\alpha\}$ and $1\{P \leq \alpha\}$ are generally different, except for the $\{0, 1/\alpha\}$ -valued $E = 1_{P \leq \alpha}/\alpha$ (in this case, the e-value depends on α)

Example

- $Y_i \sim \mathcal{N}(\mu, 1)$
- Test $H_0 : \mu = 0$ (vs $H_1 : \mu > 0$)
- Let $Z_n = \sqrt{n}\bar{Y}_n$
- p-variables

$$P_n = \Phi(-Z_n)$$

- e-variables

$$E_n = e^{Z_n - 1/2}$$

- $\mathbb{P}(P_n \leq \alpha) = \alpha$
- $\mathbb{P}(E_n \geq 1/\alpha) = \mathbb{P}(Z_n \geq 1/2 - \log \alpha) = 1 - \Phi(1/2 - \log \alpha)$

α	Z_n thr. (e-test)	Type-I error (e-test)	Z_n thr. (p-test)	Type-I error (p-test)
0.10	2.80	2.5e-3	1.28	0.10
0.05	3.49	2.4e-4	1.64	0.05
0.01	5.10	1.6e-7	2.32	0.01

2.2 Using distributional knowledge

- $1/\alpha$ cut-off always works for any E variable
- If the null distribution of E is known/computable, use its survival function $S(x; \mathbb{P}) = \mathbb{P}(E \geq x)$ to get a smaller threshold t^* than $1/\alpha$:

$$t^* = \inf\{t > 0 : \sup_{\mathbb{P} \in \mathcal{P}} S(t; \mathbb{P}) \leq \alpha\}.$$

- Essentially similar strategy to finding threshold in Neyman-Pearson test
- In our example, we find $t^* = \exp(-\Phi^{-1}(\alpha) - 1/2)$; test with this threshold is identical to the p-test

2.3 Calibrators: converting between p- and e-values

Calibrators

- *p-to-e*: $f : [0, \infty) \rightarrow [0, \infty]$ decreasing with $f = 0$ on $(1, \infty)$; $f(P)$ is an e-variable for any p-variable P .
- f is a *p-to-e* calibrator $\iff \int_0^1 f(p)dp \leq 1$
- *e-to-p*: $f : [0, \infty) \rightarrow [0, \infty)$ decreasing; $f(E)$ is a p-variable for any e-variable E .
- “Admissible” (when restricting to $[0,1]$ p-variable) e-to-p: $f(t) = \min(1, 1/t)$.
- Useful p-to-e choices: $2(1 - p)$, $p^{-1/2} - 1$, $-\log p$, and power families $\kappa p^{\kappa-1}$, $\kappa \in (0, 1)$.
- Iterating both directions is wasteful

2.4 Randomized tests & randomized e-to-p

Randomized Markov

If $Y \geq 0$ and $U \sim \text{Unif}(0, 1)$ independent of Y , then for any $\alpha > 0$,

$$\mathbb{P}(Y \geq U/\alpha) \leq \alpha \mathbb{E}[Y].$$

- For an e-variable E , the randomized test $1\{E \geq U/\alpha\}$ is (often) more powerful than $1\{E \geq 1/\alpha\}$.
- If P is a p-variable independent of E , then P/E is a p-variable (randomized e-to-p).
- In particular, U/E is a p-variable

2.4 Randomized tests & randomized e-to-p

Example

- $E_n = e^{Z_n - 1/2}$
- Randomized test: reject with probability $\min(1, \alpha E_n)$
- $\tilde{P}_n = U/E_n$ with $U \sim U(0, 1)$

$$\begin{aligned}\mathbb{P}(\tilde{P}_n \leq \alpha) &= \mathbb{E}^{\mathbb{P}}[\Pr(U \leq \alpha E_n \mid E_n)] \\ &= \mathbb{E}^{\mathbb{P}}[\min(1, \alpha E_n)] \\ &= \alpha \Phi(-1/2 - \log \alpha) + \Phi(\log \alpha - 1/2)\end{aligned}$$

α	Type-I error
0.05	0.049922
0.01	0.009999

2.5 Admissible e-variables

- Intuition: E is *admissible* if it cannot be uniformly improved (no dominating e-variable).
- **Simple nulls:** E is admissible iff exact. Any exact E equals an LR dQ/dP for some $Q \ll P$.
- **General nulls:** characterization; simple sufficient conditions (e.g., exactness).

Corollary (Sufficient admissibility conditions)

An e-variable E for $\mathcal{P}$ is admissible if any of the following conditions holds:

- (a) for each $A \in \mathcal{F}$ with $\mathcal{P}_A \neq \emptyset$, there exists $P \in \mathcal{P}_A$ such that $\mathbb{E}^P[E] = 1$;
- (b) for each $P' \in \mathcal{P}$, there exists $P \in \mathcal{P}$ such that $P' \ll P$ and $\mathbb{E}^P[E] = 1$;
- (c) E is exact.

2.6 Examples

Monotone likelihood ratio (composite and parametric null)

- $(\mathbb{P}_\theta)_{\theta \in \Theta}$ parametric family of distributions with densities p_θ of the sufficient statistics Z
- **Monotone likelihood ratio** holds: for any $\theta < \theta'$, $p_{\theta'}(z)/p_\theta(z)$ exists and is increasing in z
- E.g. $\mathcal{N}(\theta, \sigma^2)$, one-parameter exponential family
- Null $\mathcal{P} = \{\mathbb{P}_\theta \mid \theta \leq \theta_0\}$
- For any $\theta' > \theta_0$,

$$E = \frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)}$$

is an e-variable for $\mathcal{P}$.

2.6 Examples

Monotone likelihood ratio (composite, parametric hypothesis)

Proof.

Write $\mathbb{E}^\theta$ for $\mathbb{E}^{\mathbb{P}_\theta}$. For any $\theta \leq \theta_0 < \theta'$, we have

$$\begin{aligned} 1 &= \mathbb{E}^{\theta_0} \left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)} \right] = \mathbb{E}^\theta \left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)} \frac{p_{\theta_0}(Z)}{p_\theta(Z)} \right] \\ &\geq \mathbb{E}^\theta \left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)} \right] \mathbb{E}^\theta \left[\frac{p_{\theta_0}(Z)}{p_\theta(Z)} \right] = \mathbb{E}^\theta \left[\frac{p_{\theta'}(Z)}{p_{\theta_0}(Z)} \right], \end{aligned}$$

where the inequality follows from Chebyshev's association inequality. This gives $\mathbb{E}^\theta[E] \leq 1$ for all $\theta \leq \theta_0$, as required. $\square$

Chebyshev's association inequality: If f and g are nondecreasing, then $\mathbb{E}[f(X)g(X)] \geq \mathbb{E}[f(X)]\mathbb{E}[g(X)]$.

2.6 Examples

Testing symmetry (composite and nonparametric null)

- $Z \in \mathbb{R}$, $\mathcal{P} = \{\mathbb{P} \mid Z \sim \mathbb{P} \Rightarrow -Z \sim \mathbb{P}\}$
- Let $q(z) > 0$ be some pdf and define

$$p^*(z) = \frac{1}{2}(q(z) + q(-z)).$$

- Then

$$E = \frac{q(Z)}{p^*(Z)} = \frac{2q(Z)}{q(Z) + q(-Z)}$$

is an exact e-variable for $\mathcal{P}$.

Proof.

$$\begin{aligned}\mathbb{E}^{\mathbb{P}}[E] &= \mathbb{E}^{\mathbb{P}} \left[\frac{q(Z)}{q(Z) + q(-Z)} \right] + \mathbb{E}^{\mathbb{P}} \left[\frac{q(Z)}{q(Z) + q(-Z)} \right] \\ &= \mathbb{E}^{\mathbb{P}} \left[\frac{q(Z)}{q(Z) + q(-Z)} \right] + \mathbb{E}^{\mathbb{P}} \left[\frac{q(-Z)}{q(Z) + q(-Z)} \right] = 1.\end{aligned}$$



2.6 Examples

Testing group invariance (composite and nonparametric null)

- Finite group G
- $Z \in \mathbb{R}^p$, $\mathcal{P} = \{\mathbb{P} \mid Z \stackrel{d}{=} g(Z) \text{ under } \mathbb{P} \text{ for all } g \in G\}$
- Examples: flip sign $G = \{g(z), g(-z)\}$, exchangeability group $G = \{g(Z) = Z_\sigma, \sigma \in \mathcal{S}_p\}$
- Let $q(z) > 0$ be some pdf and define

$$p^*(z) = \frac{1}{|G|} \sum_{g \in G} q(g(z))$$

- Then

$$E = \frac{q(Z)}{p^*(Z)} = \frac{|G|q(Z)}{\sum_{g \in G} q(g(Z))}$$

is an exact e-variable for $\mathcal{P}$.

2.6 Examples

Testing the mean (composite and nonparametric null)

- $Z \in [0, \infty)$
- $\mathcal{P} = \{\mathbb{P} \mid \mathbb{E}^{\mathbb{P}}[Z] \leq \mu\}$ (mean at most μ)
- (alternative: mean larger than μ)
- For any $\lambda \in [0, 1]$,

$$E = (1 - \lambda) + \lambda \frac{Z}{\mu}$$

is an (admissible) e-variable for $\mathcal{P}$

- If now $\mathcal{P} = \{\mathbb{P} \mid \mathbb{E}^{\mathbb{P}}[Z] \leq \mu, \text{var}^{\mathbb{P}}(Z) \leq \sigma^2\}$ (mean at most μ , variance at most σ^2)
- For any $\lambda \in [0, 1]$,

$$E = (1 - \lambda) + \lambda \frac{Z^2}{\mu^2 + \sigma^2}$$

is an e-variable for $\mathcal{P}$

2.7 Interpreting realized e-values: informal grids

Rule-of-thumb by analogy with Jeffreys' scale for Bayes factors:

e-value	evidence
$[0, 1]$	supports H_0
$(1, 3.16)$	no more than a bare mention
$[3.16, 10)$	substantial
$[10, 31.6)$	strong
$[31.6, 100)$	very strong
$[100, \infty)$	decisive